Curriculum Learning, Image Robustness, and Region-Focused Preprocessing for Hierarchical Dental Disease Detection in the DENTEX 2023 Panoramic X-ray Dataset

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Abstract

Panoramic dental X-rays are one of the most common imaging modalities in clinical dentistry, and automated detection of abnormal teeth could reduce the burden of manual review. The DENTEX 2023 benchmark released a hierarchically annotated collection of 2,032 labeled panoramic X-rays covering quadrant, tooth-enumeration, and disease-diagnosis labels, but most prior work has trained only on the 705 images in the fully labeled tier, discarding roughly two-thirds of the available supervision. Using the full DENTEX 2023 release, we study whether a three-stage curriculum that moves from quadrant localization, to tooth enumeration, and finally to disease detection improves a YOLOv8m segmentation model relative to a single-stage baseline trained only on the fully labeled tier. On the 250-image DENTEX test set, the curriculum model reaches box and segmentation mAP@0.5 of 0.394 and 0.386, compared with 0.417 and 0.400 for the baseline. This is a small but consistent gap, and it holds in the same direction across the validation set as well. *[Placeholder for RQs 3 and 4]* Together, these results suggest that on datasets of this scale do not automatically improve fine-grained detection. Class imbalance and the intrinsic difficulty of particular diseases are larger limits than the total number of labeled boxes.

1 Introduction and Literature Review

Dental disease is one of the most common chronic conditions worldwide, and panoramic radiography is the standard first-line imaging technique used to screen for it (Peres et al. 2019). A single panoramic X-ray typically contains 28 to 32 teeth, and clinicians must assess each of them individually for decay, periapical lesions, and other abnormalities. The task is repetitive, and the volume of panoramic X-rays taken in routine care is substantial. These two features together make panoramic-X-ray reading a natural target for automation.

Modern object-detection models have made rapid progress on this kind of problem. The YOLO family in particular has become the default baseline for medical-image detection because it is fast, well supported in open-source tooling, and produces both bounding-box and polygon-mask outputs in a single pass (Jocher et al. 2023). Recent work in dentistry has applied YOLO-family detectors to caries detection, periapical-lesion detection, and tooth-numbering tasks on both bitewing and panoramic X-rays (Hamamci et al. 2023a). A recurring concern is that dental disease classes are heavily imbalanced in naturally occurring data, and that fully labeled panoramic-X-ray datasets remain small.

The DENTEX 2023 benchmark, released as part of the MICCAI 2023 challenge (Hamamci et

al. 2023a, Hamamci et al. 2023b), was designed in part to address this data-scarcity problem by providing annotations at three levels of detail. The release includes 693 X-rays labeled at the quadrant level only, 634 X-rays labeled with quadrant plus tooth position within the quadrant, and 705 X-rays labeled with full diagnosis codes. The coarser labels encode useful structure. A model that first learns to localize quadrants should have an easier time learning tooth positions, and a model that has learned tooth positions should be better placed to assign disease classes to the correct tooth. The general idea that easier-to-harder tasks can improve a model's final performance has a long history in machine learning, going back to the original formulation of curriculum learning (Bengio et al. 2009).

Two concerns naturally follow. First, whether an easy-to-hard curriculum on DENTEX actually improves disease-level detection compared with a matched single-stage baseline that uses the fully labeled tier alone. If the extra supervision in the first two stages does not carry classification power to the final task, then the additional labeled images are not functioning as a useful prior, and the curriculum is largely overhead. Second, whether the curriculum's contribution is concentrated in a particular stage. If most of the useful representation is learned in the quadrant stage for example, that has different implications for dataset design than if most of it is learned in the enumeration stage.

This paper addresses those concerns directly. We train a three-stage curriculum YOLOv8m segmentation model on the DENTEX 2023 release and compare it, under matched hyperparameters, with a single-stage baseline trained only on the fully labeled tier. We also examine the checkpoints produced at each stage of the curriculum to isolate which stage contributes the most to the final model. A parallel track of the project evaluates how the resulting model behaves under realistic image degradation and across different scales of anatomical input.

The four research questions are as follows:

1. *RQ1.* Does a three-stage curriculum (quadrant, enumeration, diagnosis) improve DENTEX disease-detection performance over a single-stage baseline trained only on the fully labeled tier?
2. *RQ2.* Which curriculum stage contributes most to the final model's performance?
3. *RQ3.* To what extent do realistic image degradations such as blur, noise, and motion reduce detection performance, and can augmentation with these degradations improve robustness?
4. *RQ4.* To what extent does region-focused preprocessing at different anatomical scales improve detection performance relative to using the full panoramic image?

The primary goal of this project is empirical rather than methodological. We do not propose a new architecture or a new curriculum algorithm. Instead, we evaluate how well a standard, widely used detection model responds to an explicit easy-to-hard curriculum when the curriculum is built using data that has already been released alongside the target task.

2 Data Collection

2.1 Source and scope

All data comes from the DENTEX 2023 release (Hamamci et al. 2023a), a panoramic-X-ray benchmark assembled from three different clinical institutions and annotated under the FDI (International Dental Federation) tooth-numbering system. DENTEX is distributed as a single archive

containing four sub-directories that correspond to its annotation tiers, each with its own annotation format and image pool.

- Tier 1 contains 693 panoramic X-rays with quadrant-only labels (classes 0–3, corresponding to FDI quadrants 1–4).
- Tier 2 contains 634 panoramic X-rays labeled with quadrant plus tooth position within the quadrant.
- Tier 3 holds 705 panoramic X-rays labeled with quadrant, tooth position, diagnosis class, and segmentation mask.
- Tier 4 has 1,571 panoramic X-rays with no annotations.

Tiers 1 and 2 are distributed in COCO format with one JSON file per tier. Tier 3 is also COCO-formatted but uses three separate category fields to encode quadrant, tooth, and diagnosis respectively, and includes segmentation information.

We used the full Tiers 1–3 collection for training and validation, and the 250-image test set for final evaluation.

2.2 Annotation structure and target classes

The final modeling target is the four DENTEX disease classes. These are Impacted (obstructed tooth eruption), Caries (early decay, cavities), Periapical Lesion (infection at the root tip), and Deep Caries (advanced decay, cavities approaching the pulp of the tooth).

For the curriculum stages, two intermediate target sets are used. Stage 1 targets the four quadrants. Stage 2 targets the eight tooth positions within a quadrant. Stage 3 targets the four disease classes above. The hierarchical relationship between the three target sets is exactly what motivates the curriculum. A detection at Stage 3 corresponds to a specific tooth within a specific quadrant carrying a specific diagnosis, and each of those three pieces of information is supervised separately at some point in the pipeline.

For convenience, Table 1 summarizes the three curriculum stages together with the single-stage baseline referenced throughout the rest of the paper.

Table 1: Quick reference for the four training runs evaluated in this paper. All four produce YOLOv8m-seg checkpoints. Stage 3 (Curriculum) and Baseline are directly compared in RQ1, and all three curriculum stages are compared against each other in RQ2.

Run	Training images	Classes	Warm-start	Role
Stage 1 (Quadrant)	1,398	4 (quadrants)	ImageNet	Curriculum foundation
Stage 2 (Enumeration)	1,339	8 (tooth positions)	Stage 1 best	Curriculum middle
Stage 3 (Curriculum)	705	4 (diseases)	Stage 2 best	Curriculum final
Baseline	705	4 (diseases)	ImageNet	Comparison for RQ1

2.3 Preprocessing and conversion to YOLO format

Raw DENTEX annotations are not in a format YOLO can train on directly, so the first preprocessing step converts them. Three preprocessing decisions deserve explicit attention. First, polygon segmentation masks were preserved for Stage 3, so the final curriculum stage and the baseline both produce segmentation outputs in addition to bounding boxes. This matches the DENTEX test-set annotation format, which also includes polygon masks. Second, for Stages 1 and 2, the Tier 3 images carry coarser quadrant- and tooth-level labels in addition to their disease labels, so they can be folded into the earlier stages by extracting only the coarser labels. This is what allows Stage 1 and Stage 2 to see roughly twice as much training data as a naive single-stage model would. Third, the 131 unmappable test-set annotations mentioned above were dropped during conversion rather than being silently assigned to a default class.

3 Exploratory Data Analysis

Before training any detectors, we first examined the training data to understand its class composition, spatial structure, and object-size distribution. The patterns that emerge from that exploration motivate several of the modeling choices described in the following section.

3.1 Class distribution

At the disease level, the four DENTEX training classes are imbalanced. Across the 3,529 annotations in the 705 Stage 3 training images (of which 678 carried at least one labeled disease instance), Caries is by far the most common class. Impacted is the next most common and corresponds to a visually very distinct shape (third molars that fail to erupt). Deep Caries appears less often, and Periapical Lesion is rare. The frequency ratio between the most and least common classes is roughly 14 to 1 (Figure 1). This imbalance is more severe than the class ratios alone suggest, because the rare classes also tend to be smaller objects and to appear in fewer spatial positions, both of which compound the difficulty of learning them from a fixed number of training boxes.

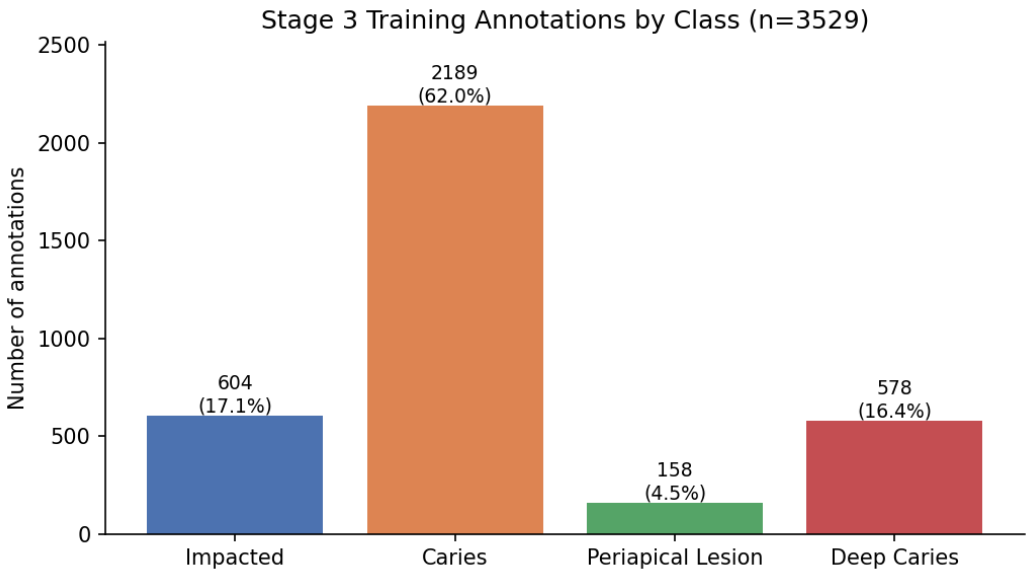

Figure 1: Per-class annotation counts for the 3,529 disease-level annotations in the 705 Stage 3 training images. Caries dominates, and Deep Caries accounts for less than one-tenth of the combined Caries plus Deep Caries count.

The class distribution at the quadrant and enumeration stages is much more uniform. Every image contains multiple quadrants and multiple teeth, so the per-image labeling for Stages 1 and 2 is an order of magnitude larger than for Stage 3. This asymmetry is an important feature of the curriculum. The earlier stages are not only on larger image pools, they also produce more labels per image, which drives a much stronger and more stable training signal at those stages than at the final one.

3.2 Bounding-box size

Bounding boxes in DENTEX are small relative to the image, with a mean area of roughly 1.3% of the full panoramic view (Figure 2). The distribution is right-skewed. Most teeth occupy only a small fraction of the image, and the largest boxes correspond to impacted third molars with surrounding structures or to diseases that span multiple adjacent teeth. Impacted boxes are on average slightly larger than Caries or Deep Caries boxes, consistent with the fact that impacted third molars typically occupy more image area than a decayed surface on a single tooth.

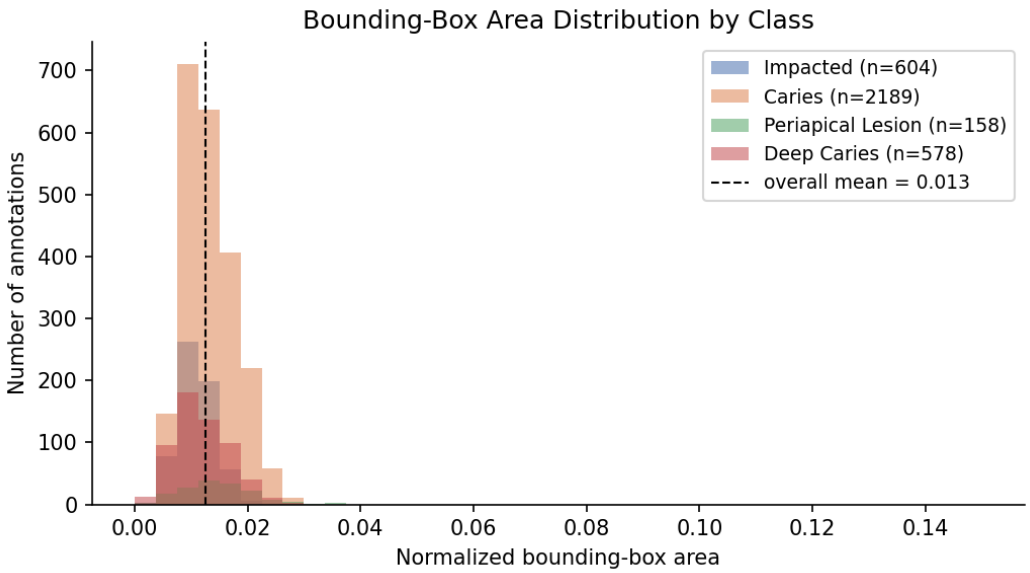

Figure 2: Normalized bounding-box area distribution by class. The mean area over all annotations is approximately 0.013 of the full image. The distribution is right-skewed with a long but thin tail.

3.3 Spatial structure

The spatial distribution of annotated objects in the training set reflects the anatomical structure of a panoramic radiograph. Annotations cluster in the left and right regions of the image (corresponding to the left and right dental arches) and are sparse near the vertical midline (Figure 3). Within each side, detections are distributed along a smooth arc. This is the structural feature that makes region-focused preprocessing worth evaluating in RQ4.

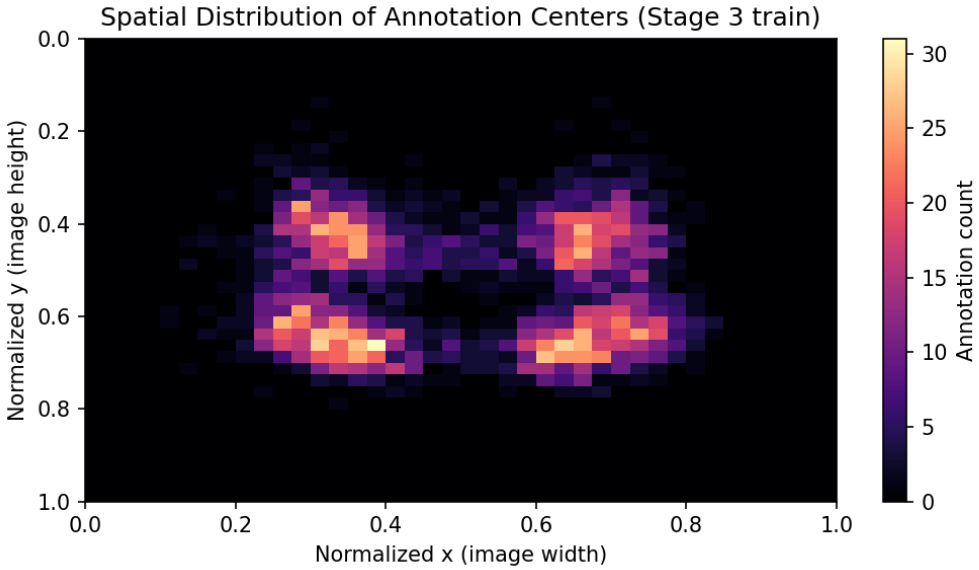

Figure 3: Two-dimensional density of annotation centers in normalized image coordinates. Two bright lobes correspond to the left and right dental arches, and the central region between them is effectively empty.

3.4 Objects per image

The number of disease annotations per image is also right-skewed, with a mean of 5.2 disease instances per annotated image, a median of 4, and a maximum of 23 (Figure 4). Stage 3 thus presents the model with many small annotated regions scattered against a much larger unlabeled background, a setting that detectors are known to find difficult (Lin et al. 2017). Datasets at this scale typically benefit more from stronger augmentation than from architectural changes.

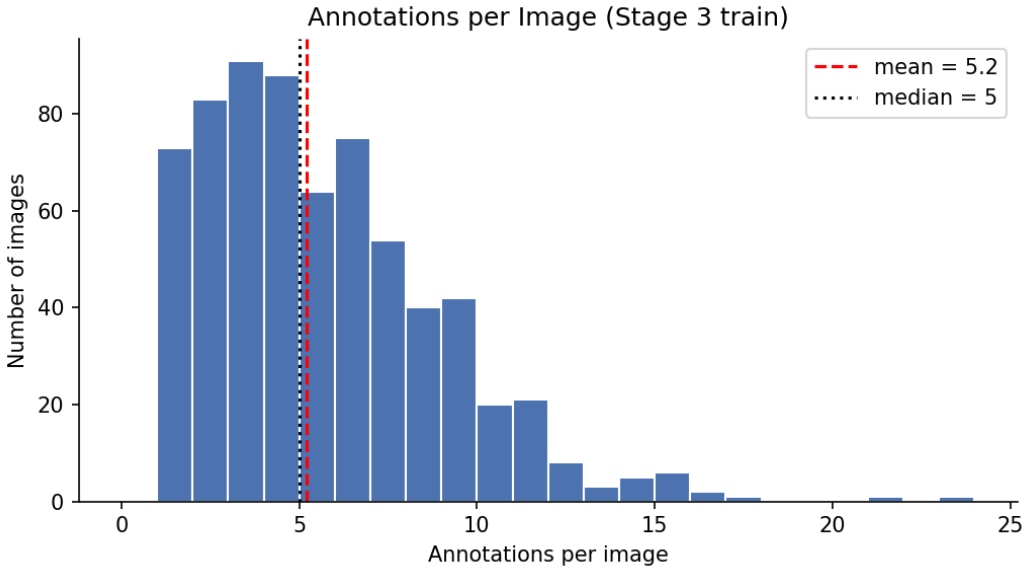

Figure 4: Distribution of annotated disease instances per image across the 678 annotated Stage 3 training images. Mean 5.2, median 4, maximum 23.

4 Methods

4.1 Model

The underlying detector for all runs is YOLOv8m-seg (Jocher et al. 2023), the medium-sized variant of the YOLOv8 family. It produces both bounding boxes and polygon segmentation masks in a single forward pass, which matches the format of the DENTEX test-set annotations. All four runs (Stage 1, Stage 2, Stage 3 Curriculum, and Baseline) use the same architecture and start from the same ImageNet-pretrained weights.

4.2 Curriculum training framework (RQ1 and RQ2)

The curriculum is implemented as a sequence of three fine-tuning runs in which the best checkpoint from the previous stage becomes the initialization for the next. All three stages share the same optimizer family, cosine learning-rate schedule, weight decay, and full-strength mosaic augmentation. They differ primarily in their initial learning rate, warm-up length, and input data.

- Stage 1 (Quadrant detection). Initialized from ImageNet. Trains on 1,398 images (693 Tier-1 plus 705 Tier-3 at the quadrant level), targets four quadrant classes, with moderate augmentation.
- Stage 2 (Tooth enumeration). Initialized from the best Stage 1 checkpoint. Trains on 1,339 images (634 Tier-2 plus 705 Tier-3 at the enumeration level), targets eight tooth-position classes, at a lower learning rate than Stage 1.
- Stage 3 Curriculum (Diagnosis). Initialized from the best Stage 2 checkpoint. Trains on the 705 Tier-3 images, targets four diagnosis classes, at the lowest learning rate of the three stages and with slightly stronger augmentation than Stage 1 to compensate for the smaller training pool.

All four runs use early stopping on validation mAP and a fixed upper bound on training epochs.

4.3 Baseline

The baseline for RQ1 is a single-stage run matched as closely as possible to Stage 3 so that any performance difference can be attributed to the presence of the curriculum warm-start rather than to hyperparameter drift. The baseline trains on the same 705 Tier-3 images, with the same architecture, optimizer, and augmentation schedule as Stage 3. The two differences are the initialization (ImageNet instead of Stage 2's checkpoint) and a slightly higher initial learning rate, since Stage 3 uses a lower one as the final fine-tuning stage of a warm-started model. Other optimizer choices are identical across the two runs.

4.4 *[Placeholder for methods for RQ3 and 4]*

4.5 Evaluation

All curriculum and baseline models are evaluated on the held-out DENTEX validation set (50 panoramic X-rays) during training and on the DENTEX test set (250 panoramic X-rays, 1,469 evaluable annotations after filtering) for final reporting. The validation set is used for picking the best checkpoint inside each training run. The test set is used only once, after all hyperparameters and stopping criteria have been fixed, to avoid leaking test-set information into model selection.

Two standard metrics are reported on both sets. Mean average precision at IoU 0.5 (mAP@0.5) is the headline metric used in the DENTEX challenge and summarizes performance at a single, relatively forgiving overlap threshold. We also report mAP@0.5-0.95, which averages performance across stricter overlap thresholds (from 0.5 up to 0.95) and is more sensitive to how tightly each prediction is localized. Both metrics are reported for bounding boxes and for segmentation masks. We treat mAP@0.5 as the primary metric because it is most directly comparable to the DENTEX challenge baselines and is more stable on a small test set. For RQ2, which compares the three curriculum stages against each other, we also report each stage's validation mAP on its own target (quadrants for Stage 1, tooth positions for Stage 2, diseases for Stage 3). These per-stage values are not directly comparable to one another, since they measure different tasks, but the overall profile of how mAP develops across stages is informative about where the curriculum is doing its work.

5 Results

5.1 Sample description

The final working corpus comprised 1,398 training images at Stage 1, 1,339 at Stage 2, and 705 at Stage 3, with 50 validation images and 250 test images used identically across runs. Within the 705 fully labeled Tier-3 training images, 678 carried at least one disease annotation, for a total of 3,529 disease instances in the Stage 3 and baseline training pool (mean 5.2 annotations per annotated image). After removing the 131 unmappable test annotations, the test set contained 1,469 evaluable disease instances distributed across the four DENTEX classes.

Figure 5 reports validation mAP@0.5 per training epoch for all three comparable runs (Baseline, Stage 2, and Stage 3 Curriculum). The three trajectories converge on a similar ceiling for box detection, and the Stage 3 curriculum run does not overshoot the baseline's final mAP at any point. This visual already previews the RQ1 result reported below.

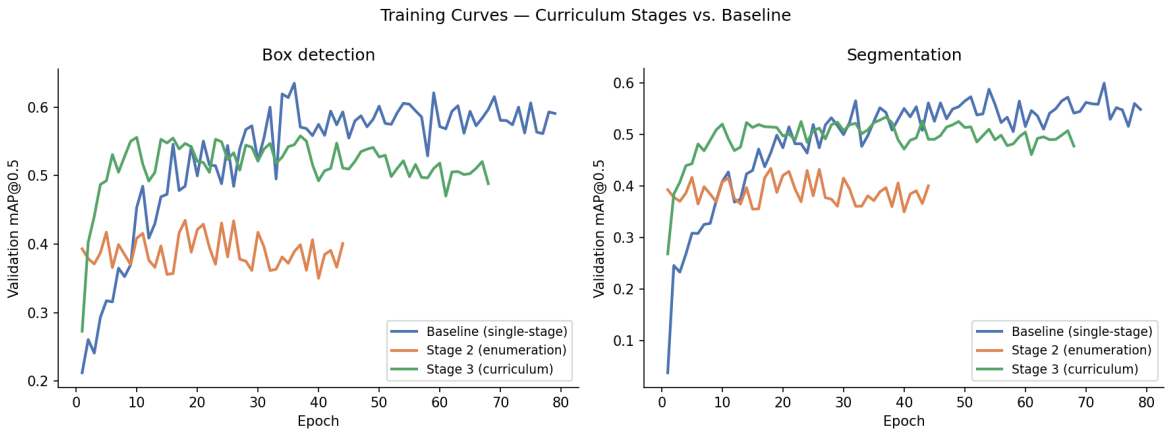

Figure 5: Validation mAP@0.5 per training epoch for the three comparable runs. Stage 2 targets eight tooth-position classes, while Stage 3 and Baseline target the four disease classes, so only the Stage 3 vs Baseline comparison is direct. Stage 2 is shown to illustrate the relative training-signal strength of the enumeration stage.

5.2 RQ1. Curriculum versus single-stage baseline

The headline comparison is between the final curriculum model (Stage 3, warm-started from Stage 2) and the matched single-stage baseline (started from ImageNet weights). Both are trained on the same 705 Tier-3 disease images, share the same architecture, optimizer, and augmentation schedule, and are evaluated identically on the DENTEX 50-image validation set (used for selecting the best checkpoint) and the 250-image test set (used once, at the end). Table 2 reports the best-epoch values for each model.

Table 2: Validation and test performance for the final curriculum stage (Stage 3) and the matched single-stage baseline, best-epoch values on the 50-image DENTEX validation set and the 250-image test set. All four disease classes scored jointly. The baseline is ahead of the curriculum on every metric and at both IoU thresholds.

Model	Val box mAP@0.5	Val seg mAP@0.5	Val box mAP@0.5-0.95	Val seg mAP@0.5-0.95	Test box mAP@0.5	Test seg mAP@0.5
Baseline (single-stage)	0.635	0.600	0.415	0.374	0.417	0.400
Stage 3 (Curriculum)	0.558	0.534	0.371	0.355	0.394	0.386
Δ (Baseline − Curriculum)	+0.077	+0.066	+0.044	+0.019	+0.023	+0.014

The baseline is ahead of the curriculum on every metric in Table 2 (Figure 6). The test-set gap is modest in absolute terms (+0.023 box mAP@0.5, +0.014 seg mAP@0.5), but the direction is

consistent across all six matched comparisons (box and segmentation, at both overlap thresholds, on validation and on test). That uniformity is not compatible with the two models being equivalent on this data. The gap also widens at the stricter overlap threshold on the validation set (+0.044 box), which suggests the baseline is also producing tighter, better-localized predictions and not just finding more of the right objects.

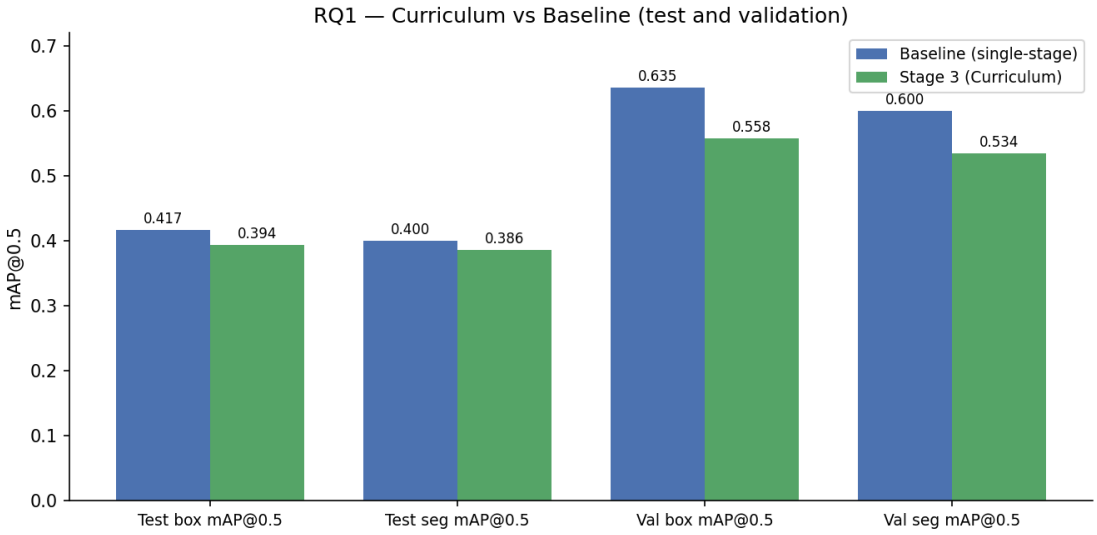

Figure 6: Test-set and validation mAP@0.5 for curriculum versus baseline, for both box and segmentation outputs. The baseline is ahead across every matched comparison, and the val-set gap (about 0.07 mAP) is larger than the test-set gap (about 0.02 mAP).

The curriculum does inherit a real but short-lived advantage from the Stage 2 warm-start. At the first training epoch it reports 0.273 val box mAP@0.5 against the baseline's 0.212, a +0.061 head start that reflects the Stage 2 model having already learned to recognize tooth-shaped regions. That advantage does not persist. By the best-val epoch (36 for baseline, 37 for curriculum, nearly identical), the baseline has overtaken the curriculum by 0.077 val box mAP and ends at a higher ceiling under every overlap threshold evaluated. The curriculum is also more expensive to train: the baseline reaches its best epoch in roughly half the time that Stages 2 and 3 together require, before Stage 1 is even counted.

Two factors likely explain this outcome. First, the early stages teach the model essentially geometric information (where each quadrant is, where each tooth is) rather than what diseases look like. So the extra supervision the curriculum supplies is mostly orthogonal to what the final disease task needs. Second, because the set of output classes changes at every transition ($4 \rightarrow 8 \rightarrow 4$), the final classification layer has to be reinitialized each time. Only the shared feature-extraction layers actually carry over, and the Stage 3 classifier ends up being learned from scratch on the same 705-image label pool the baseline sees. The one small advantage of the curriculum is that it loses less performance from validation to test (a 29% relative decline versus 34% for the baseline), which suggests the curriculum representation generalizes very slightly better. But the baseline's absolute numbers are still higher, so this does not change the practical picture.

5.3 RQ2. Which stage contributes most

The curriculum is a sequence of three training phases, each with its own target space, training-image pool, and annotation density. RQ2 asks where, if anywhere, the useful part of the curriculum's representation is being built. Two breakdowns are needed to answer that, because the three stages are not on a shared mAP scale. Table 3 reports the stages' peak validation mAPs on their own target spaces, and Table 4 reports the training-signal volume (total number of annotated boxes) each stage exposes the model to.

Table 3: Best validation mAP@0.5 (box) at each stage on that stage's own target space, plus the baseline for context. These per-stage values are *not* directly comparable, because each stage is predicting a different thing over a different number of classes. The correct reading is the overall *profile* rather than a ranking. Stage 1's per-epoch metrics were not archived in the recovered project state, so they are not reported here.

Stage	Target space	Train images	Best val mAP@0.5 (box)	Peak epoch
Stage 1	Quadrant (4 classes)	1,398	not archived	not archived
Stage 2	Tooth enumeration (8 classes)	1,339	0.435	18 of 44
Stage 3 (Curriculum)	Disease (4 classes)	705	0.558	37 of 68
Baseline	Disease (4 classes)	705	0.635	36 of 79

Table 4: Training-signal volume at each stage of the curriculum, measured directly from the per-stage label files. The curriculum's total training-signal volume (Stages 1 + 2 + 3 = 31,454 boxes) is 8.9× the volume the baseline sees from its 3,529 Tier-3 boxes, but most of that extra volume (21,624 boxes, or 69% of the curriculum total) comes from Stage 2.

Stage	Annotated images	Boxes	Boxes per annotated image	Class imbalance (max to min)
Stage 1	1,371	6,301	4.6	1.1 to 1 (very balanced)
Stage 2	1,312	21,624	16.5	1.3 to 1 (balanced)
Stage 3 (Curriculum / Baseline)	678	3,529	5.2	13.9 to 1 (severely imbalanced)

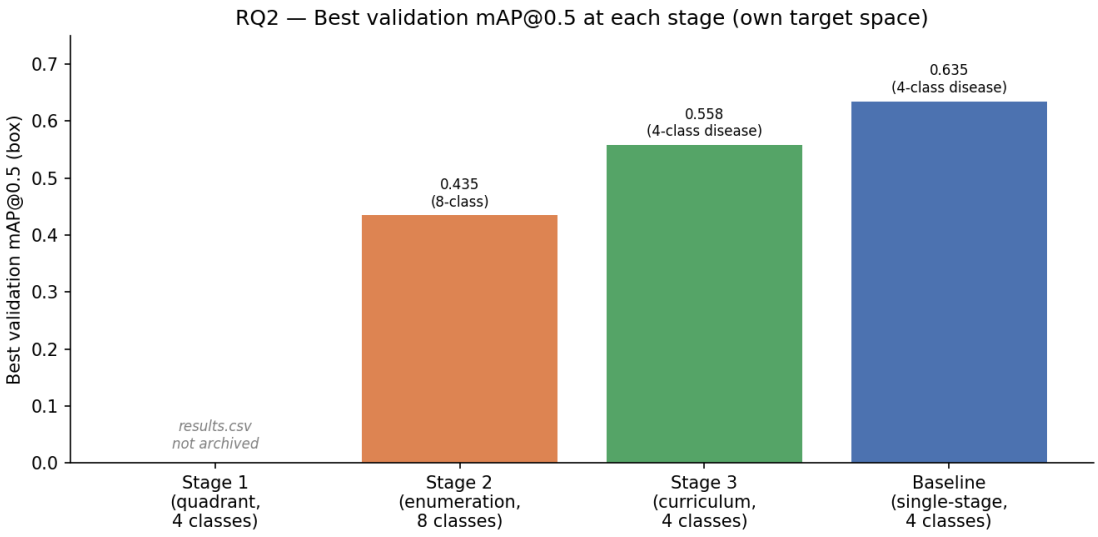

Figure 7: Best validation mAP@0.5 (box) at each stage of the curriculum, each on its own target space, plus the baseline (single-stage) for comparison. Targets differ across stages (quadrant, tooth, disease), so absolute heights are *not* directly comparable. The figure is informative about the *profile* of the curriculum, not a ranking of which stage is "best".

Most of the curriculum's extra supervision is concentrated at Stage 2, which sees 21,624 annotated tooth-position boxes. That is 6.1× what Stage 3 sees and 3.4× what Stage 1 sees, accounting for 69% of the curriculum's 31,454-box total. Stage 1 is spatially coarse (roughly one box per dental quadrant) and Stage 3 has only its own small training pool, so any useful head start the curriculum hands to Stage 3 has to come mostly from Stage 2. This matches the first-epoch observation from RQ1: the curriculum begins training at 0.273 val mAP because Stage 2 has already taught the model to recognize tooth-shaped regions.

The extra supervision still does not translate into a higher ceiling. Stage 3 (curriculum) peaks at 0.558 val box mAP@0.5 on the four-class disease task, while the baseline peaks at 0.635 on the same task from only an ImageNet start. Even with 8.9× more annotated boxes across the full curriculum, the warm-started model ends up below the cold-started model trained on 705 images alone. The apparent drop from Stage 2 (0.435) to Stage 3 (0.558) in Table 3 is not evidence that the curriculum is forgetting what it learned. Stage 2 is predicting eight tooth-position classes with 16.5 boxes per image, while Stage 3 is predicting four disease classes with 5.2 boxes per image and a 13.9-to-1 class imbalance, so the two numbers measure different tasks of different difficulty and cannot be compared directly.

Read together, the two tables suggest the curriculum is putting most of its effort into the wrong problem for the downstream task. Stage 2 carries most of the box volume and most of the representation-building, but its target (tooth identity under FDI numbering) has the least to do with the Stage 3 disease target. The final classification layer is reinitialized at each transition ($4 \rightarrow 8 \rightarrow 4$ with different class meanings), so only the shared feature layers carry forward. The Stage 3 classifier still has to be learned from scratch on the same 705-image, 13.9-to-1 imbalanced label pool the baseline faces, which is the actual bottleneck in both runs.

5.4 Per-class performance on the test set

Table 5 decomposes the curriculum's test-set box mAP@0.5 by diagnosis class. The difference between classes is larger than the difference between the curriculum and baseline models.

Table 5: Per-class test-set AP@0.5 (box) for the curriculum model. Impacted is nearly at the ceiling, and Deep Caries is essentially not learned at this sample size.

Class	Test AP@0.5 (box)
Impacted	0.942
Caries	0.421
Periapical Lesion	0.193
Deep Caries	0.021

Impacted teeth are easy. They are visually distinctive (unerupted third molars with a characteristic orientation), they tend to appear in predictable locations, and they are well represented in training. At the other extreme, Deep Caries is close to a failure case. It is the rarest class in training, and its appearance overlaps substantially with the more common Caries class, from which it is normally told apart only by how deeply the decay has reached into the pulp. With so few training examples, the rarer classes are effectively crowded out by the more common ones. Caries and Periapical Lesion fall between these two extremes. The overall mAP@0.5 of 0.394 is, in that sense, an average over one nearly-solved class, two partially-solved classes, and one almost-unsolved class.

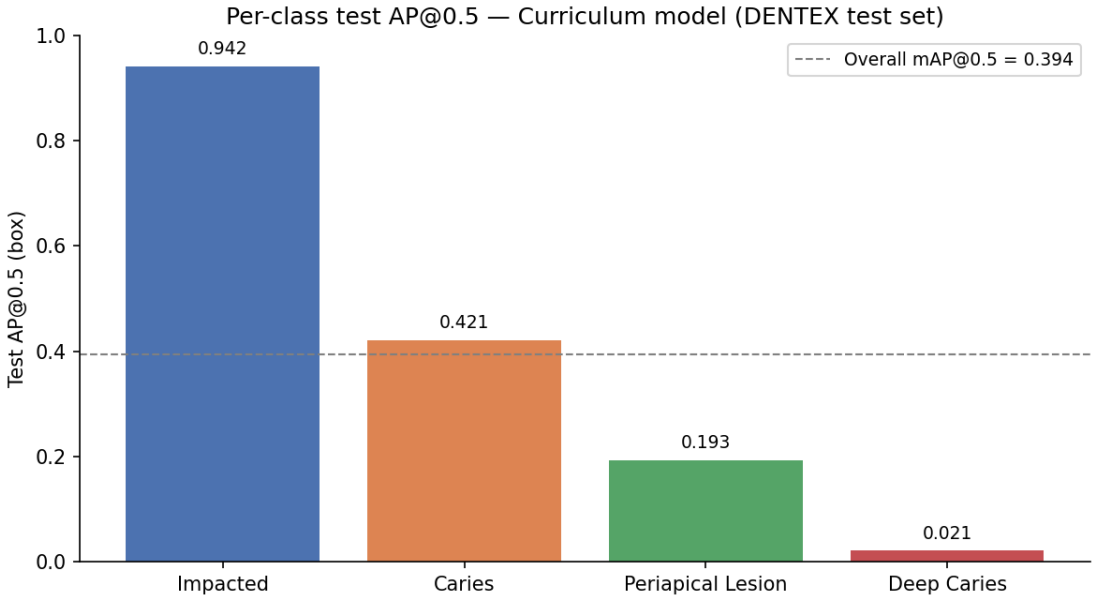

Figure 8: Per-class AP@0.5 on the 250-image DENTEX test set for the curriculum model. The spread between classes is an order of magnitude larger than the spread between models, with Impacted nearly at the ceiling and Deep Caries close to zero in both models.

This per-class profile (Figure 8) is also relevant to RQ1. Because the curriculum-versus-baseline gap is small in absolute terms (0.02 mAP), and because the per-class spread is an order of magnitude

larger, any intervention that improves Deep Caries or Periapical Lesion by even a modest amount would produce a larger overall change than the curriculum intervention itself. This is a useful ranking for deciding where to spend additional effort.

5.5 Confusion structure

The AP numbers collapse each class's behavior into a single scalar, so we also look at the validation-time normalized confusion matrices for the two models (Figure 9). The matrices make the trade-off in the curriculum run more legible than the AP table does. Under both models, the Impacted class is close to the ceiling. The true-positive rate sits at 0.90, with the remaining mass going to background rather than to another disease class. The curriculum run nudges the Caries recall upward (0.47 → 0.51) and reduces the rate at which background is labeled as caries (0.49 → 0.42). But that improvement comes at a cost on the two rarer lesion classes. The Periapical Lesion true-positive rate falls from 0.44 in the baseline to 0.22 in the curriculum run, and Deep Caries falls from 0.50 to 0.34. In both models, the Caries, Deep Caries, and Periapical Lesion block is substantially off-diagonal: when the model does flag one of these classes, it is nearly as likely to confuse it with another caries-like class as to label it correctly. The curriculum does not eliminate that swapping. If anything, it shifts recall toward the most common disease class at the expense of the two rarer ones. This is consistent with the explanation given in the Discussion: the representation inherited from the 8-class tooth-enumeration task helps for classes that are well-represented in training (Impacted, Caries), but the fine-grained distinctions needed for the rarer lesions still have to be learned at Stage 3 on 705 images, which is the same constraint the baseline faces.

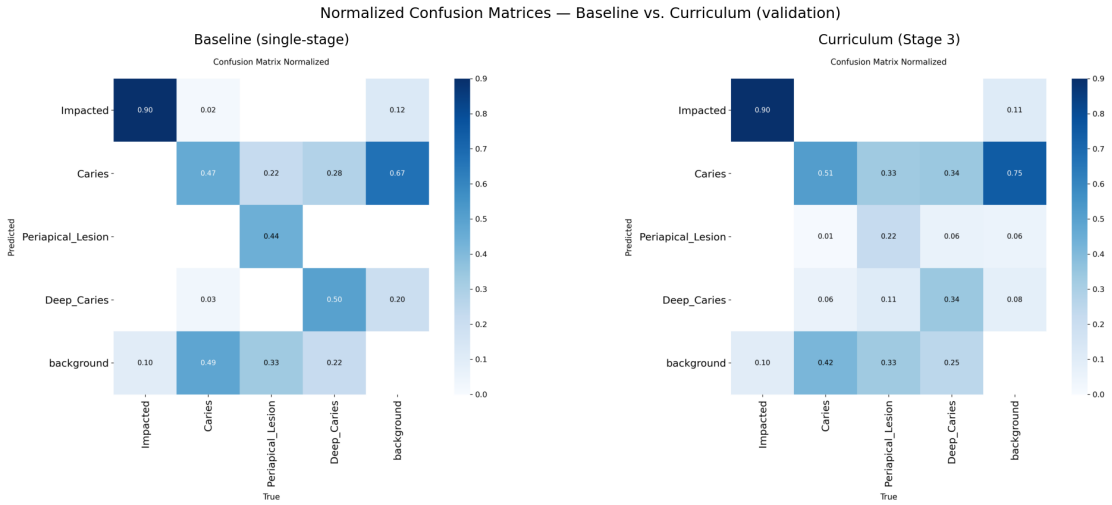

Figure 9: Normalized confusion matrices on the validation set for the single-stage baseline (left) and the full curriculum run (right). Both models are near the ceiling on Impacted (0.90). The curriculum trades a small improvement on Caries (0.47 → 0.51) for a sharp regression on Periapical Lesion (0.44 → 0.22) and a moderate regression on Deep Caries (0.50 → 0.34). Off-diagonal mass among the three caries-like classes is substantial in both runs.

5.6 Qualitative predictions

Figure 10 shows six representative test-set predictions from the curriculum model. The common failure modes visible in these examples match the quantitative profile above. Impacted third molars

are reliably localized and classified. Superficial caries on anterior teeth are usually detected but sometimes misclassified between Caries and Deep Caries. Periapical lesions near root tips are sometimes missed entirely when the surrounding bone texture is low-contrast. No sample was selected adversarially, and the six shown are a stratified draw across the four classes.

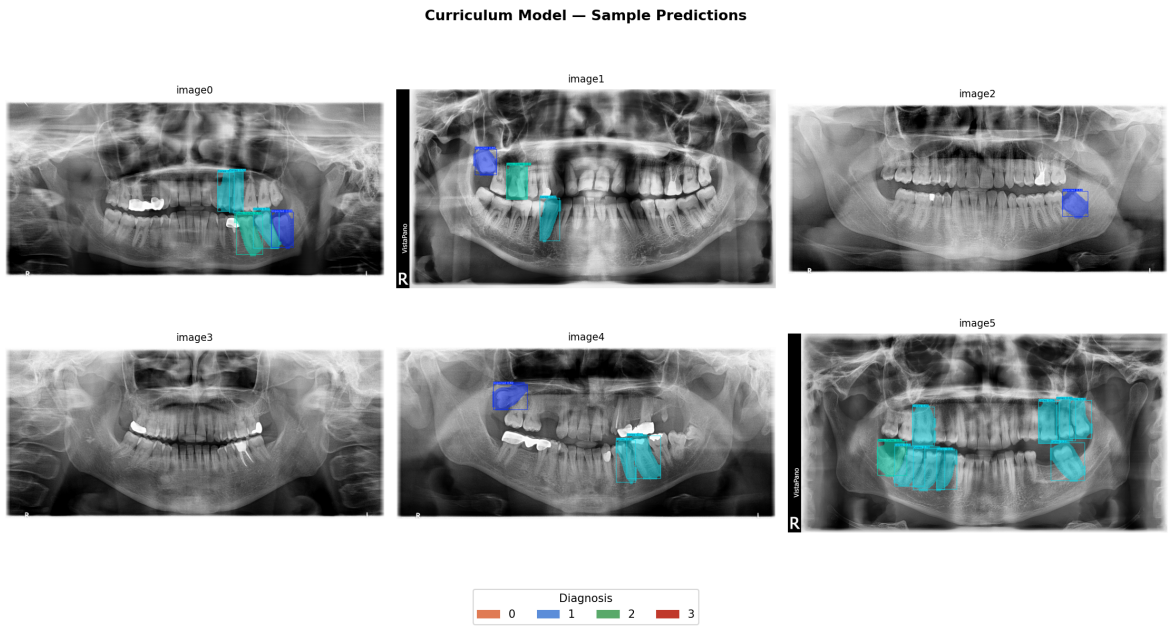

Figure 10: Six representative test-set predictions from the curriculum model, with bounding boxes and class labels overlaid. Failure modes match the per-class profile. Impacted is reliable, Caries versus Deep Caries is the dominant confusion, and periapical lesions are occasionally missed.

5.7 [Placeholder for RQ3]

5.8 [Placeholder for RQ4]

5.9 Cross-task consistency

One of the clearer findings across both RQ1 and RQ2 is that the most important limits on final performance are not the ones targeted by the curriculum. Whether the model is warm-started from ImageNet or from Stage 2, its ranking of class difficulty is essentially unchanged. Impacted is easy, Deep Caries is hard, and the middle two are middle. This mirrors the pattern reported for the DENTEX baseline method (Hamamci et al. 2023b) and suggests that at the current sample size, the dominant source of error is not the quality of the shared representation but the label sparsity of individual disease classes.

6 Discussion

Under matched hyperparameters, the three-stage curriculum did not improve disease detection on DENTEX, and the negative direction is consistent across all six matched comparisons (test and validation, box and segmentation, both overlap thresholds). That the baseline also reaches its best checkpoint in a shorter total schedule, and at roughly half the total training time of the full curriculum, reinforces the point. The most plausible reading is that both models hit a ceiling set by

the 705-image, 13.9-to-1 imbalanced Tier-3 training pool, and the curriculum's extra supervision does not move that ceiling upward despite giving the model a measurable head start at the very first epoch.

The mechanism behind that outcome is visible in the per-stage breakdown. Stage 2 contributes 21,624 of the curriculum's 31,454 annotated training boxes (69% of the total) and is the last representation-shaping step before the diagnosis task begins. But Stage 2's target is tooth position under FDI numbering, which is a geometric label rather than a pathology one. The curriculum therefore concentrates most of its training effort on the stage whose labels are least predictive of the eventual disease classes. Training-signal volume is not the real bottleneck. The baseline sees 8.9× less labeled data and still wins. What limits Stage 3 is the distribution of disease labels: four classes with a 13.9-to-1 ratio between most and least common, and a rare class (Deep Caries, 158 boxes) whose appearance overlaps with a common class (Caries, 2,189 boxes). No amount of Stage 1 or Stage 2 supervision relaxes that. Consistent with this, the per-class spread on the test set (0.94 for Impacted down to 0.02 for Deep Caries) is an order of magnitude larger than the curriculum-versus-baseline gap (about 0.02), which suggests that future work on DENTEX will be better served by class-level interventions (class-balanced sampling, weighted losses, or synthetic augmentation of the rare classes) than by further curriculum engineering.

Limitations. We did not repeat training runs to estimate run-to-run noise directly. Any single one of the six matched comparisons in isolation would be compatible with noise, and the strength of the RQ1 conclusion therefore rests on the consistency of direction across comparisons rather than on a formal statistical test. All curriculum runs also share a single backbone and a single family of hyperparameters, so a curriculum that does not help under these settings might still help under a different model family or augmentation recipe.

7 Conclusion

Three findings stand out from the curriculum-learning track. First, a three-stage easy-to-hard curriculum trained on the full DENTEX 2023 release did not outperform a matched single-stage baseline. The baseline was ahead across all six matched comparisons (test and validation, box and segmentation, both overlap thresholds), with the test-set box mAP@0.5 gap at about +0.023 in the baseline's favor. Second, the curriculum does inherit a short-lived warm-start advantage at the very first epoch from the Stage 2 model, but that advantage is spent by the best-validation epoch (around epoch 36 for both models), after which the curriculum settles at a lower ceiling. Third, most of the curriculum's extra annotated-box volume is concentrated at Stage 2, whose tooth-position target overlaps spatially with disease classification but not semantically, so the additional supervision is plentiful but largely off-target for what the final classifier needs. Taken together, these results suggest that on datasets of DENTEX's scale, class imbalance and the intrinsic difficulty of particular diseases are larger limits than the total number of labeled boxes, and that curricula built from whatever label tiers a dataset happens to ship with do not automatically improve fine-grained detection.

The robustness (RQ3) and region-focused-preprocessing (RQ4) tracks of this project are reported separately by the co-author and provide complementary evidence on how the model behaves under controlled distribution shift and across different anatomical scales of input.

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